Film Festival Database
Relational Model and Relational Algebra

1 Relational Model

Festival(id, title, year, location)

Category(id, name)

Person(id, name, type)

Staff(id, name, type, specialization)

Film(id, title, duration, year, country, category_id, final_score)

Screening(id, date, time, room, capacity, film_id, coordinator_id)

FestivalCategory(festival_id, category_id)

FestivalFilm(festival_id, film_id)

FilmDirector(film_id, director_id)

FilmActor(film_id, actor_id)

Evaluation(judge_id, film_id, score)

2 Relational Algebra Queries

1. Films in a specific festival

$$\pi_{\text{Film.*}}\left(\text{Film} \bowtie_{\text{Film.id=FestivalFilm.film\_id}} \left(\sigma_{\text{festival\_id}=x}(\text{FestivalFilm})\right)\right)$$

2. Number of films in each category

$$_{\text{id, name}}\Im_{\text{COUNT Film.id AS film\_count}}\left(\text{Category} ⟕_{\text{Category.id=Film.category\_id}} \text{Film}\right)$$

3. Directors of a film

$$\pi_{\text{name}}\left(\text{Person} \bowtie_{\text{Person.id=FilmDirector.director\_id}} \sigma_{\text{film\_id}=x}(\text{FilmDirector})\right)$$

4. Actors of a film

$$\pi_{\text{name}}\left(\text{Person} \bowtie_{\text{Person.id=FilmActor.actor\_id}} \sigma_{\text{film\_id}=x}(\text{FilmActor})\right)$$

5. Future screenings of a film

$$\begin{aligned}\pi_{\text{title, time, date, room, capacity, name}}\Big(&\sigma_{\text{film\_id}=x\ \wedge\ \text{datetime>now}}(\text{Screening})\\ &\bowtie_{\text{Screening.coordinator\_id=Staff.id}} \text{Staff}\\ &\bowtie_{\text{Screening.film\_id=Film.id}} \text{Film}\Big)\end{aligned}$$

6. Number of screenings for each film

$$\begin{aligned}\mathcal{T}_{\text{screening\_count DESC, title}}\Big(&{}_{\text{id, title}}\Im_{\text{COUNT Screening.id AS screening\_count}}\Big(\\ &\text{Film} ⟕_{\text{Film.id=Screening.film\_id}} \text{Screening}\Big)\Big)\end{aligned}$$

7. Average score for each film

$$_{\text{id, title}}\Im_{\text{ROUND(AVERAGE score, 2) AS average\_score}}\left(\text{Film} \bowtie_{\text{Film.id=Evaluation.film\_id}} \text{Evaluation}\right)$$

8. Films ranked by average score

$$\begin{aligned}\mathcal{T}_{\text{average\_score DESC, title}}\Big(&{}_{\text{id, title}}\Im_{\text{ROUND(AVERAGE score, 2) AS average\_score}}\Big(\\ &\text{Film} \bowtie_{\text{Film.id=Evaluation.film\_id}} \text{Evaluation}\Big)\Big)\end{aligned}$$

9. Films without evaluations

$$\begin{aligned}\pi_{\text{Film.id, Film.title}}\Big(&\sigma_{\text{Evaluation.film\_id IS NULL}}\big(\\ &\text{Film} ⟕_{\text{Film.id=Evaluation.film\_id}} \text{Evaluation}\big)\Big)\end{aligned}$$

10. Judges with more than five evaluations

$$\tau_{\text{films\_evaluated DESC}}\Big(\sigma_{\text{films\_evaluated}>5}\Big(\text{id, name, specialization}\\ \Im_{\text{COUNT film\_id AS films\_evaluated}}\Big(\\ \sigma_{\text{type='judge'}}(\text{Staff})\\ \bowtie_{\text{Staff.id=Evaluation.judge\_id}} \text{Evaluation}\Big)\Big)\Big)$$

11. Films with multiple directors

$$\sigma_{\text{director\_count}>1}\Big(\text{id, title } \Im_{\text{COUNT director\_id AS director\_count}}\Big(\\ \text{Film} \bowtie_{\text{Film.id=FilmDirector.film\_id}} \text{FilmDirector}\Big)\Big)$$

12. Actors appearing in more than two films

$$\sigma_{\text{film\_count}>2}\Big(\text{id, name } \Im_{\text{COUNT actor\_id AS film\_count}}\Big(\\ \text{Person} \bowtie_{\text{Person.id=FilmActor.actor\_id}} \text{FilmActor}\Big)\Big)$$

13. Screenings on a specific date

$$\pi_{\text{title, time, room, capacity, name}}\Big(\sigma_{\text{date}=x}(\text{Screening})\\ \bowtie_{\text{Screening.coordinator\_id=Staff.id}} \text{Staff}\\ \bowtie_{\text{Screening.film\_id=Film.id}} \text{Film}\Big)$$

14. Number of screenings in each room on a specific date

$$\text{room } \Im_{\text{COUNT * AS screening\_count}} \left(\sigma_{\text{date}=x}(\text{Screening})\right)$$

15. Judges without evaluations

$$\pi_{\text{Staff.id, Staff.name, Staff.specialization}}\Big(\tau_{\text{Staff.name}}\big(\\ \sigma_{\text{judge\_id IS NULL}}\big(\\ \sigma_{\text{type='judge'}}(\text{Staff})\\ ⟕_{\text{Staff.id=Evaluation.judge\_id}} \text{Evaluation}\big)\big)\Big)$$

16. Films without screenings

$$\pi_{\text{Film.id, Film.title}}\Big(\sigma_{\text{Screening.film\_id IS NULL}}\big(\\ \text{Film} ⟕_{\text{Film.id=Screening.film\_id}} \text{Screening}\big)\Big)$$

17. Screenings by coordinator

$$\pi_{\text{id, date, time, room, capacity, title}}\Big(\sigma_{\text{coordinator\_id}=x}(\text{Screening}) \\ \bowtie_{\text{Screening.film\_id}=\text{Film.id}} \text{Film}\Big)$$

18. Films evaluated by a specific judge

$$\pi_{\text{Film.id, Film.title, Evaluation.score}}\Big(\sigma_{\text{judge\_id}=x}\big(\\ \text{Evaluation} \bowtie_{\text{Evaluation.film\_id}=\text{Film.id}} \text{Film}\big)\Big)$$

19. Director info for a specific film

$$\pi_{\text{Person.*}}\Big(\text{Person} \bowtie_{\text{Person.id}=\text{FilmDirector.director\_id}} \big(\\ \text{FilmDirector} \bowtie_{\text{FilmDirector.film\_id}=\text{Film.id}} \sigma_{\text{Film.id}=x}(\text{Film})\big)\Big)$$